RED SEAL EXAM PREP

456A Welder

Free 50-Question Mock Exam

Practice for the Red Seal / Certificate of Qualification exam

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Mock Exam — answer all questions, then check the key

Q1. **Which filler metal is used for GTAW of chrome-moly (Cr-Mo) steel such as P91?**

A) ER308L

B) ER70S-2

C) ER90S-B9

D) ER4043

Q2. **During GMAW the arc pops and stutters, the wire occasionally burns back to the contact tip, and the bead comes out irregular. What is the most likely cause?**

A) Work angle too steep for the joint

B) Worn drive rolls or a kinked liner

C) Travel speed too slow for the amperage

D) Shielding gas flow rate set too low

Q3. **Once their package has been opened, low-hydrogen electrodes (E7018) must be stored in:**

A) An oven held at the electrode maker's temperature

B) A freezer, so that the flux coating cannot oxidize

C) Any dry spot in the shop, out of direct sunlight

D) A sealed plastic bag kept at room temperature

Q4. **E6013 is preferred for welding thin sheet metal because:**

A) It provides maximum penetration into thin material

B) It produces a hard slag that prevents porosity

C) It has a soft arc, easy slag removal, and works on AC or DC

D) It generates the highest amperage range

Q5. **Which type of fire extinguisher is suitable for a welding-area electrical fire?**

A) Class C (dry chemical)

B) Class D (dry powder)

C) Class B (foam type)

D) Class A (water type)

Q6. **A fillet weld symbol on a shop drawing shows 3-12 to the right of the weld symbol, with dimensions in inches. How much unwelded length is left between two consecutive weld segments on that side of the joint?**

A) 9 in

B) 12 in

C) 6 in

D) 15 in

Q7. What is the function of anti-spatter spray in GMAW?

A) To reduce oxidation of the weld pool

B) To lubricate the wire liner

C) To improve shielding gas coverage

D) To keep spatter from sticking to the nozzle

Q8. What causes a "crater crack" at the end of a SMAW weld?

A) Rapid contraction of the pool at arc break

B) Using DCEN instead of DCEP

C) Moisture in the low-hydrogen electrode coating

D) Excessive amperage throughout the weld

Q9. Excessive spatter in FCAW-G is most likely caused by:

A) Wire stickout too short

B) Shielding gas flow rate too high

C) Arc voltage set too low

D) Base metal temperature too high

Q10. Which GMAW transfer mode requires the highest arc voltage to achieve?

A) Short circuit transfer

B) Pulsed spray transfer

C) Globular transfer

D) Spray transfer

Q11. In the FCAW electrode classification E71T-1, what do the "T" and the trailing "-1" designate?

A) T = travel direction; -1 = flat position only

B) T = tubular electrode; -1 = single-pass use only

C) T = titanium flux coating; -1 = wire diameter

D) T = tubular electrode; -1 = CO_2 or Ar/CO_2, DCEP

Q12. Welding on a container that previously held flammable liquid is safe when:

A) The container has been left open to air out for a full 24 hours

B) The welder works quickly so as to minimize the heat input

C) The container is purged, tested and certified gas free

D) The liquid has been removed and the container rinsed with water

Q13. What is the leg of a fillet weld?

A) The height of the weld reinforcement above the base metal

B) The distance from the weld root to the toe

C) The total length of the fillet weld deposit

D) The total depth of penetration into the base metal

Q14. **What is the neutral flame setting in oxyfuel welding?**

A) Equal volumes of oxygen and acetylene

B) Oxygen valve fully open, acetylene barely open

C) Excess oxygen, no feather

D) Excess acetylene, large feather present

Q15. **ER70S-6 wire has higher silicon and manganese content than ER70S-3. Why?**

A) To allow welding with 100% CO_2 shielding gas only

B) To enable AC power source compatibility

C) To add deoxidizers for welding slightly rusty surfaces

D) To increase tensile strength above 80 ksi

Q16. **What is plasma arc cutting (PAC) best suited for compared to OFC?**

A) Cutting materials that do not conduct electricity

B) Cutting galvanized steel outdoors

C) Cutting stainless steel and aluminum

D) Cutting thick carbon steel over 100 mm

Q17. **What is globular transfer in GMAW and when does it occur?**

A) Transfer of large droplets at intermediate voltage — high spatter

B) Transfer of very fine droplets at high current — preferred for production welding

C) Transfer where the wire dips into the molten pool at low voltage

D) Transfer mode unique to pulsed GMAW at low background current

Q18. **What is the difference between a groove weld and a fillet weld?**

A) Groove welds are partial penetration only; fillet welds are always full penetration

B) Groove welds fill a prepared groove; fillet welds join surfaces at an angle

C) Groove welds are used for lap joints; fillet welds are used for butt joints

D) Groove welds require GTAW; fillet welds can use SMAW only

Q19. **What is the purpose of preheating steel before SMAW?**

A) To improve slag fluidity for easier removal

B) To increase arc voltage

C) To slow the cooling rate of the joint

D) To increase the deposition rate

Q20. **What is the purpose of the high-frequency (HF) start in GTAW?**

A) To generate the shielding gas ionization

B) To start the arc without touching the workpiece

C) To increase amperage for thicker material

D) To provide the cleaning action for aluminum

Q21. **What does E7018 mean in the AWS electrode classification system?**

A) 70 ksi tensile, flat/horizontal only, rutile, AC only

B) 7,000 psi tensile, all-position, cellulosic, DCEP

C) 70 ksi tensile, vertical-down only, iron powder

D) 70 ksi tensile, all-position, low-hydrogen, DCEP/AC

Q22. **What does "walking the cup" technique mean in GTAW?**

A) Bracing the nozzle cup against the base metal and rocking it to move the torch steadily

B) Oscillating the filler rod in and out of the leading edge of the puddle to control bead width

C) Moving the torch in a wide circular pattern so the heat spreads evenly across the joint and the puddle stays fluid

D) Whipping the torch back and forth to let the puddle cool between advances

Q23. **What is the purpose of the liner in a GMAW torch?**

A) To regulate shielding gas flow

B) To insulate the torch from the work cable

C) To guide the wire to the contact tip

D) To cool the contact tip

Q24. **What is the main characteristic of an E6010 electrode?**

A) Rutile coating, easy slag removal, AC or DC

B) High cellulosic coating, deep penetration, DCEP only

C) Iron powder coating, high deposition rate

D) Low hydrogen, requires dry storage

Q25. **What causes "sugaring" (oxidation/discolouration) on the back side of a stainless steel GTAW root pass?**

A) Lack of back purge on the root side

B) Excessive heat input from high amperage

C) Insufficient post-flow gas time

D) Using the wrong filler wire for the base metal

Q26. **Under AWS D1.1, a welder passes a groove weld qualification test on plate in the 3G (vertical) position. Which plate groove positions does this qualify them to weld?**

A) 1G, 2G, and 3G

B) All positions including 4G and 6G

C) 3G only

D) 1G and 2G only

Q27. **What is "burn-back" in GMAW?**

A) Porosity from shielding gas loss

B) The wire burning back to the contact tip

C) Undercutting at high wire feed speeds

D) Excessive spatter on the base metal

Q28. **What is sensitization in austenitic stainless steel?**

A) Martensite formation from rapid quenching

B) Surface oxidation from insufficient shielding gas

C) Chromium carbide precipitation at grain boundaries

D) Excessive heat input causing grain growth

Q29. **A welder notices the electrode holder cable insulation is cracked. The correct action is:**

A) Continue welding if the crack is small

B) Replace the cable before continuing work

C) Wrap the crack with electrical tape and continue

D) Reduce amperage to lower risk

Q30. **What does the "hot work permit" system control?**

A) Authorization to weld in fire-risk areas

B) Maximum cylinder storage pressure limits

C) Electrode preheat and interpass temperatures

D) The maximum amperage allowed in a shop

Q31. **What type of tungsten tip preparation is used for GTAW on steel with DCEN?**

A) A split (notched) tip

B) A sharp tapered point

C) A rounded ball tip

D) A flat-ground tip

Q32. **Why is AC polarity used for GTAW of aluminum?**

A) AC lowers total heat input so thin aluminum will not burn through

B) AC gives deeper penetration into aluminum than straight polarity DCEN

C) AC prevents tungsten contamination

D) The EP half-cycle removes the aluminum oxide layer

Q33. **Undercut in SMAW is caused by:**

A) Moisture in the flux coating

B) Insufficient preheat of the base metal

C) Too low an amperage setting

D) Excessive arc length or amperage

Q34. **What is the maximum working (delivery) pressure allowed for acetylene at the torch?**

A) 207 kPa (30 psi)

B) 345 kPa (50 psi)

C) 103 kPa (15 psi)

D) 690 kPa (100 psi)

Q35. **A 456A welder is welding mild steel at a fixed indoor workstation. Under CSA W117.2 and provincial occupational health and safety regulation, what determines whether the ventilation is adequate?**

A) 1,000 cfm of general dilution air per welder whenever the shop ceiling is under 5 m

B) Nothing below 100 A — ventilation applies only to high-current welding on coated metal

C) Fume kept below the occupational exposure limits, with local exhaust at the source

D) 2,000 cfm of general dilution air per welder, the fixed minimum set by Canadian regulation

Q36. **Why must oxygen hoses and fittings be kept free of oil and grease?**

A) Grease reduces gas flow rate

B) Oil causes corrosion of brass fittings

C) They can ignite spontaneously in pure oxygen

D) Oil contaminates the flame and changes its characteristics

Q37. **Which shielding gas mixture is most commonly used for GMAW of mild steel?**

A) 50% Argon / 50% Helium

B) 100% carbon dioxide

C) 100% Argon

D) 75% Argon / 25% CO_2

Q38. **What is the difference between a Welding Procedure Specification (WPS) and a Procedure Qualification Record (PQR)?**

A) WPS is the test result; PQR is the written instruction

B) They are interchangeable documents

C) PQR applies to welder qualification; WPS applies to procedure qualification

D) WPS is the written instruction; PQR is the qualifying test record

Q39. **What is the gas lens in a GTAW torch?**

A) A gas pressure regulator built into the torch body

B) A glass lens that focuses the shielding gas into a beam

C) A lens that focuses the arc onto a smaller area for precision

D) A mesh screen that produces smooth, laminar gas flow

Q40. **What is the approximate spray transition current for ER70S-6 wire (0.9 mm / 0.035") with 98% argon / 2% oxygen shielding gas?**

A) Approximately 230–250 A

B) Approximately 80–100 A

C) Approximately 165 A

D) Spray transfer is possible with any gas above 200 A

Q41. What polarity is typically used for FCAW-G?

A) AC (alternating current)

B) DCEN (electrode negative)

C) DCEP (electrode positive)

D) DCEN for indoor, DCEP for outdoor

Q42. Which destructive test method provides information on the ductility and soundness of a weld root?

A) Charpy V-notch impact test

B) Guided bend test

C) Tensile test

D) Hardness test

Q43. What do AWS welding symbols mainly show?

A) The welding process and shielding gas to use

B) The type, size, and location of a weld on a drawing

C) The brand of welding machine to use

D) The electrode classification and storage requirements

Q44. A welder is welding in a low-lying area and notices dizziness. The most likely hazard is:

A) Nitrogen narcosis from shielding gas

B) Carbon monoxide from the electrode coating

C) UV radiation causing nausea

D) Oxygen displacement by argon or CO_2

Q45. What colour coding identifies a pure tungsten electrode (EWP)?

A) Green

B) Gold

C) Red

D) Grey

Q46. What does CTWD stand for in GMAW?

A) Contact Tip Wear Detection

B) Contact Tip to Work Distance

C) Circuit Travel Wire Direction

D) Current to Wire Diameter ratio

Q47. What type of slag system does E71T-1 flux-cored wire use?

A) Iron powder — high deposition rate for flat/horizontal only

B) Cellulosic slag — deep penetrating, used outdoors

C) Basic slag — used for positional welding with DCEN

D) Rutile slag — fluid, easily removed, all-position

Q48. **What is the purpose of a Charpy V-notch impact test?**

A) To measure the toughness of the weld at a set temperature

B) To measure the hardness of the HAZ

C) To measure the tensile strength of the weld metal

D) To measure porosity in the weld cross-section

Q49. **A welder is air carbon arc gouging (CAC-A) by hand with the electrode diameter and amperage already set. The groove is coming out shallower than needed. Which change will make the gouge deeper?**

A) Pull the electrode back to lengthen the arc

B) Speed up the travel speed along the groove

C) Slow down the travel speed along the groove

D) Aim the air jet at one side of the groove

Q50. **What does WHMIS stand for?**

A) Workshop Handling & Machinery Inspection Safety

B) Workplace Hazardous Materials Information System

C) Workplace Hazard Management and Inspection Sheet

D) Welding Health and Material Inspection Standards

Answer Key & Explanations

> Scoring: 35 correct out of 50 is 70%. These 50 questions come from the 125-question 456A bank; the last page shows what the complete printable bank contains.

Q1. Answer: C

P91 (9% Cr, 1% Mo) creep-resistant steel requires matching filler metal ER90S-B9 (or equivalent). Standard carbon steel or stainless filler would produce mismatched mechanical properties.

Q2. Answer: B

The feed path controls how smoothly the wire reaches the arc. Worn or wrongly grooved drive rolls, incorrect drive tension, a kinked or clogged liner, a worn contact tip or a tangled spool make the wire arrive in surges, so the arc lengthens and shortens, pops, and burns back. The other faults leave the feed alone and show themselves differently: low shielding gas flow shows up as porosity, too slow a travel speed as excessive buildup and cold lap, and a wrong work angle as undercut and poor bead placement. Check the feed system first — liner condition and length, drive roll size and groove type for the wire being run, drive tension, and the contact tip.

Q3. Answer: A

Once the package is open, low-hydrogen stick electrodes must be held in a holding oven at the temperature the filler metal manufacturer recommends, and the figure comes from the product, not from one universal band: Hobart's 7018 XLM data sheet gives an oven controlled at 250–300°F (about 120–150°C), while its 418 data sheet gives 220–350°F. Moisture is the biggest enemy of stick electrodes - it can cause porosity and hydrogen-induced (cold) cracking - so a shop shelf, a freezer or a plastic bag at room temperature is not storage for an opened low-hydrogen electrode. Some job codes also limit how long a package can be open, or require electrodes to be discarded after a set time even if they were kept in an oven, and an electrode that has been exposed is reconditioned exactly as the maker's data sheet says.

Q4. Answer: C

E6013 (rutile-coated) produces a soft arc, light spatter, easy slag removal, and shallow penetration — making it ideal for thin sheet metal where burn-through is the primary risk. Wrong answers: E6010 and E6011 have deep penetration and forceful arc (designed for root passes and pipe), which would easily burn through thin material. E7018 (low-hydrogen) is excellent for structural work but requires an oven and produces a broader bead; it is not the preferred choice for light gauge sheet. E6013's easy restriking and smooth bead also make it a good learning electrode, but its low penetration means it's NOT suitable for code-quality root passes or high-strength joints where full fusion is required.

Q5. Answer: A

A fire in energized electrical equipment is a Class C fire, so the extinguisher must carry a Class C rating. A dry chemical agent does not conduct, so it can be discharged onto live equipment without putting the operator into the current path. Water-type and foam agents both conduct electricity and must never be used on energized equipment, and a dry powder unit is formulated for burning metals such as magnesium or titanium, not for electrical gear. Where it can be done safely, de-energize the circuit first — once the power is off the fire is classified by whatever is actually burning.

Q6. Answer: A

Two numbers joined by a hyphen to the right of a fillet weld symbol mark an **intermittent** fillet weld. The first number is the length of each weld segment and the second is the **pitch**. Pitch is measured centre to centre from one segment to the next - Miller's guide to weld symbols calls it the centre-to-centre (or end-to-end) distance - and it is *not* the clear space between them. So 3-12 means 3 in welds at 12 in centres, and the unwelded length between segments is the pitch minus the segment length: 12 − 3 = 9 in. Answering 12 in treats the pitch as the clear gap. That reading would put the segments at 15 in centres, which is where the 15 in answer comes from, since it adds the segment length to the pitch. The 6 in answer has no basis in the notation. It is just half the pitch.

Q7. Answer: D

Anti-spatter spray is applied to nozzles and workpiece surfaces to prevent spatter from adhering and bonding. It reduces nozzle cleaning frequency and cleanup time.

Q8. Answer: A

When the arc is suddenly extinguished, the weld pool shrinks rapidly. If the crater is not filled, tensile stresses from contraction cause a star-shaped crater crack.

Q9. Answer: C

Low arc voltage relative to wire feed speed causes an unstable arc and excessive spatter in FCAW-G. Proper voltage-WFS balance is critical. Also, 100% CO_2 produces more spatter than Ar/CO_2 mixes.

Q10. Answer: D

Rank the modes by arc length and you rank them by voltage. Short circuit sits at the bottom: the arc is so short that the wire dips into the puddle and extinguishes the arc many times a second. Globular sits in the middle, with a longer arc carrying drops larger than the wire diameter across it. True axial spray sits at the top, because the arc has to be long enough and hot enough to pinch the wire tip into a fine stream of droplets smaller than the wire. Pulsed spray is the trap here: its peak current does climb above the spray transition, but the whole point of pulsing is to get spray-type droplet transfer at a lower average voltage and current, which is exactly what lets it run out of position. Voltage alone is not enough either — true spray also needs an argon-rich shielding gas, and a CO_2-rich gas will not give axial spray at any voltage setting.

Q11. Answer: D

Read E71T-1 one character at a time. E is electrode. 7 is the minimum tensile strength, 70 000 psi. Do not convert that number yourself: AWS publishes a separate metric classification, and in it the same wire is E491T-1, where 49 stands for 490 MPa. AWS A5.20/A5.20M:2005 Table 1 requires 70–95 ksi for the E7XT-1 classifications and Table 1M requires 490–670 MPa for E49XT-1, and the specification states that the two systems 'are not equivalent, each system must be used independently of the other.' The 1 that sits ahead of the T is the position digit and means all-position. T means the wire is tubular, that is flux-cored, as opposed to a solid GMAW wire. The -1 that follows the T is the usability designator: it identifies a rutile-type, fast-freezing slag system run on DCEP with CO_2 or an argon/CO_2 mix as the shielding gas. The two 1s in the designation therefore carry different meanings, and it is the trailing one that describes gas and polarity. Also keep the shielding split straight: E71T-1 requires an external shielding gas (FCAW-G), while a self-shielded wire such as E71T-11 or E71T-8 generates its own shielding from the core (FCAW-S) and runs with no gas bottle at all.

Q12. Answer: C

Containers must be thoroughly cleaned, purged, and then gas-tested and certified free of flammable vapour before any hot work — explosive residues can remain long after airing, soaked into scale, seams and sludge. Where a residual reading is tolerated at all, the hot-work limit is far tighter than the limit for cold work: Ontario's confined space guideline permits cold work under 10% LEL but hot work only under 5% LEL, and British Columbia requires flammable gases and vapours to be held below 20% of the LEL. A rinse, or a day with the lid off, proves nothing.

Q13. Answer: B

The leg of a fillet weld is the distance from the root of the joint to the toe of the weld, measured along the fusion face. For an equal-leg fillet, both legs are the same length.

Q14. Answer: A

A neutral flame has a 1:1 oxygen-to-acetylene ratio, producing a sharp bright inner cone with no feather — suitable for most welding.

Q15. Answer: C

Silicon (Si) and manganese (Mn) are deoxidizers. ER70S-6's higher Si/Mn content scavenges oxygen from the weld pool, improving weld quality on slightly rusty or dirty base metal and giving better wetting.

Q16. Answer: C

Oxy-fuel cutting works by rapid oxidation of iron, so it is confined to carbon and low-alloy steel. It cannot cut stainless steel — stainless is ferrous, but its chromium forms a refractory oxide that will not burn away — and it cannot cut aluminum or the other non-ferrous metals. Plasma arc cutting melts the metal with a high-temperature ionized gas jet and blows it out of the kerf, so it handles stainless, aluminum and copper readily. Plasma does need an electrically conductive workpiece with the work lead clamped to it, so it offers nothing on materials that do not conduct. Galvanized steel cuts perfectly well with oxy-fuel, and on heavy carbon steel plate oxy-fuel remains the practical choice, since plasma capacity falls away as thickness climbs.

Q17. Answer: A

Globular transfer occurs at intermediate voltages between short circuit and spray thresholds. Large irregular droplets — bigger than the wire diameter — form and fall into the pool, causing significant spatter. Generally avoided in production.

Q18. Answer: B

A groove weld occupies a groove or gap prepared in the base metal (butt joint). A fillet weld is a triangular cross-section weld joining surfaces at approximately 90° (T-joint, lap joint, corner joint).

Q19. Answer: C

Preheat slows the cooling rate of the HAZ, reducing hardness and the risk of hydrogen-induced cracking, especially in high-carbon or low-alloy steels.

Q20. Answer: B

HF start initiates the arc via a high-frequency spark across the gap between the electrode and workpiece — no contact required. This prevents tungsten contamination and electrode tip damage.

Q21. Answer: D

E7018 classification: E = electrode; 70 = minimum tensile strength of 70,000 psi; 1 = usable in all welding positions; 8 = low-hydrogen coating (potassium, iron powder), run on DCEP or AC. The ‘8’ (low-hydrogen) is what drives the handling rules: once the package is open, low-hydrogen electrodes are held in an oven at the temperature the electrode maker recommends (Hobart's 7018 XLM: 250–300°F), and some job codes also limit how long a package can be open and still be used — moisture in the coating puts hydrogen in the weld, and hydrogen causes hydrogen-induced (cold) cracking. Wrong answers confuse E7018 with E6010, a high-cellulose sodium electrode run on DCEP with a digging, deeply penetrating arc, or misread ‘1’ (all-position) as ‘1G’ flat-only. On Canadian work you will also see the metric form of the same classification: Hobart's 418 data sheet, for example, lists AWS A5.1 E7018 H4R alongside AWS A5.1M E4918 and CWB E4918-1-H4, where ‘49’ is the tensile strength in megapascals (490 MPa, about 70,000 psi), and the CWB's welding procedure guide points to CSA W48 or the AWS specifications for filler metal classifications.

Q22. Answer: A

"Walking the cup" means bracing the ceramic nozzle against the pipe and using a rocking or rolling motion to move the torch. This provides a stable torch angle and consistent arc length for pipe welding.

Q23. Answer: C

The liner (conduit liner) guides the wire electrode from the drive rolls through the torch body to the contact tip, preventing kinking and ensuring smooth wire feed.

Q24. Answer: B

E6010 has a high-cellulosic sodium coating that produces a forceful, deep-penetrating arc. It requires DCEP and is ideal for root passes on pipe and dirty/rusty steel.

Q25. Answer: A

"Sugaring" is a rough, oxidized surface on the weld root caused by atmospheric oxygen reacting with chromium in the hot metal, forming chromium oxides (Cr_2O_3) and depleting the corrosion-resistant chromium from the surface.

Q26. Answer: A

Under AWS D1.1, a plate groove test welded in the vertical position (3G) also covers the flat and horizontal positions - the CWB's comparison of AWS D1.1 (2015 edition) with CSA W47.1 lists 3G as qualifying 1G, 2G and 3G, plus fillets in 1F, 2F and 3F. It does not reach overhead: that is the separate 4G plate test, which qualifies 1G and 4G, and the two tests together, 3G and 4G, cover every plate position. One plate test does not cover everything; on pipe, a 6G test - the pipe fixed at a 45 degree angle, welded through every position - qualifies all the plate groove positions, and AWS describes it as qualifying virtually all other pipe positions. Read the extent of a qualification off the standard and edition you were tested to rather than off habit. In Canada, all welders working for a CWB-certified company must be qualified by the CWB Group - CSA W47.1 has the company use only tack welders, welders and welding operators qualified to the standard - and the test qualifies the welder in the process and position tested.

Q27. Answer: B

Burn-back occurs when the wire electrode burns back and fuses to the contact tip, stopping the weld. Caused by low wire feed speed relative to voltage.

Q28. Answer: C

When austenitic stainless (e.g., 304) is held in the 425–870 °C (800–1600 °F) sensitization range, carbon combines with chromium to form $Cr_{23}C_6$ at grain boundaries, depleting the surrounding area of corrosion-protective chromium and reducing corrosion resistance.

Q29. Answer: B

Damaged insulation creates electrocution and fire hazard. Equipment must be repaired or replaced before use.

Q30. Answer: A

Hot work permits authorize welding, cutting, or grinding in areas with combustible materials or fire hazards, and they set out the precautions — including a trained fire watch kept in place during the work and for at least 60 minutes after it ends.

Q31. Answer: B

For DCEN GTAW (steel, stainless, titanium), the tungsten is ground to a tapered point. For AC GTAW (aluminum), a rounded ball forms naturally at the tip.

Q32. Answer: D

Aluminum has a tenacious oxide layer (Al_2O_3, melting point 2050°C) that prevents fusion. During the electrode positive (DCEP) half of AC, electrons flow from the work to the electrode, breaking up the oxide — this is cathodic cleaning.

Q33. Answer: D

Undercut is a groove melted into the base metal along the weld toe, caused by excessive heat (high amperage or long arc) or an improper electrode angle.

Q34. Answer: C

Free acetylene gas is chemically unstable and can decompose explosively on its own, without any oxygen present, once its pressure climbs too high — so the fuel regulator is never set above 103 kPa (15 psi) gauge. Note what this limit applies to: it is the delivery pressure the regulator feeds to the torch, not the cylinder pressure. A full acetylene cylinder reads far higher, roughly 1700 kPa (250 psi), and that is safe only because the acetylene inside is dissolved in acetone held in a porous filler rather than existing as free gas. For the same reason the withdrawal rate is limited, commonly to about one-seventh of cylinder capacity per hour, so that acetone is not dragged out with the gas.

Q35. Answer: C

In Canada welding ventilation is an exposure-limit duty, not a fixed airflow number. Provincial regulation makes CSA W117.2 (Safety in welding, cutting, and allied processes) mandatory and then requires capture at source: British Columbia's OHS Regulation section 12.112 requires welding, cutting and similar processes to be carried out according to CSA W117.2, and section 12.114 requires effective local exhaust ventilation at any fixed work station to minimize worker exposure to harmful air contaminants. The test of adequacy is whether fume and gas stay below the occupational exposure limits for the metals and coatings involved — use the edition of W117.2 and the exposure limits your province adopts. The 2,000 cfm per welder figure offered here is American, not Canadian: OSHA 1910.252 sets that rate where the space is less than 10,000 cubic feet per welder or the ceiling is less than 16 feet, and it has no force here. The 1,000 cfm variant simply invents a Canadian version of that same US ceiling-height trigger. The idea that ventilation only matters above some current is false — fume is produced at every setting, and there is no amperage below which ventilation stops being required. The one Canadian airflow figure worth memorizing comes from the Canadian Centre for Occupational Health and Safety: a moveable exhaust hood should provide an air velocity of at least 100 ft/min (0.5 m/s) across the welding arc, placed as close to the work as practical. Local exhaust is mandatory regardless of general airflow when welding coated, galvanized or stainless material, where zinc oxide and hexavalent chromium exceed exposure limits even in a well-ventilated shop. AWS D1.1 and CSA W59 are fabrication codes and set no ventilation requirement.

Q36. Answer: C

Oxygen under high pressure can cause oil and grease to ignite spontaneously (oxygen-enriched combustion). This can cause violent fires or explosions.

Q37. Answer: D

C25 (75% Ar / 25% CO_2) is the most widely used mixture for mild steel GMAW. It provides a stable arc, good penetration, and lower spatter than 100% CO_2.

Q38. Answer: D

A WPS specifies the welding variables used for production welding of a joint. A PQR documents the actual test results (mechanical testing) that demonstrate the WPS produces a sound weld.

Q39. Answer: D

A gas lens replaces the standard collet body. Its porous metal mesh screen produces a smooth, laminar (non-turbulent) flow of shielding gas, extending coverage and enabling longer electrode stickout.

Q40. Answer: C

Spray transfer requires an argon-rich gas (at least ~80% Ar). With 98Ar/2O_2, 0.9 mm ER70S-6 transitions from globular to spray at roughly 165 A; with 90Ar/10CO_2 the transition rises to about 200 A. With C25 (75/25) true axial spray cannot be achieved — the high CO_2 content disrupts the arc column.

Q41. Answer: C

Gas-shielded flux-cored wire is run on DCEP, direct current electrode positive, the same polarity as GMAW solid wire. AWS A5.20/A5.20M:2005 Table 2, Electrode Usability Requirements, defines the term in a footnote as direct current electrode positive, or reverse polarity, and its polarity column is where the word "typically" in the question comes from. The gas-shielded usability designators are 1, 2, 5, 9 and 12. Designators 1, 2, 9 and 12, which include the widely used E71T-1, are listed for DCEP, and the annex describes the T-1 wires as designed for single and multiple pass welding using DCEP. Designator 5 is the exception: the flat-and-horizontal E70T-5C and E70T-5M electrodes are listed DCEP, but the all-position E71T-5C and E71T-5M electrodes are listed "DCEP or DCEN", with a footnote that some of them are recommended for DCEN to improve out-of-position welding and that the manufacturer should be consulted. Self-shielded wire is where polarity really varies: designators 3, 4 and 6 are DCEP, while 7, 8, 10, 11, 13 and 14, the majority and the common E71T-8 and E71T-11 among them, are DCEN. So a gas-shielded wire is set up electrode positive unless its classification or the manufacturer's data sheet says otherwise, and the leads are never left where the last spool had them: wrong polarity on a flux-cored wire shows up as an erratic arc, heavy spatter, porosity and poor fusion. The polarity column of that table carries only DCEP, DCEN, or "not specified" for the G classifications, so AC is not a classified polarity for any carbon steel flux-cored designator. Nor does any designator tie polarity to where the work is done: indoors or outdoors decides between gas-shielded and self-shielded wire, not between electrode positive and electrode negative.

Q42. Answer: B

The guided bend test bends a weld specimen around a mandrel to reveal cracks, porosity, and incomplete fusion, especially in the root area. Required by most qualification codes.

Q43. Answer: B

Welding symbols communicate weld type, size, length, location and finishing requirements on engineering drawings; the reference for them is AWS A2.4:2020, Standard Symbols for Welding, Brazing, and Nondestructive Examination (the latest edition listed on ANSI's webstore). Key elements: the reference line is horizontal - a weld symbol below the line calls for an arrow-side weld, above the line an other-side weld, and on both sides a weld on both sides. The tail can carry optional notes such as the process or the electrode, but those notes are supplementary, not what the symbols mainly show: Miller calls the tail the place for supplementary information on the weld, which may reference the welding process, electrode or any helpful information that does not have a place on the symbol, and Moran's text calls it an area for specifics such as the weld process, the welding procedure and material specifications. What a set of welding symbols is shorthand for, in Miller's words, is the type of weld, its size and other processing and finishing information. So a welding process with its shielding gas, or an electrode classification, can go in the tail as an optional note if it is given at all, storage requirements are not part of the symbol, and a brand of welding machine is not the type, size or location of a weld - none of the three is what the symbols mainly show. Two details to read carefully: a flag at the junction of the arrow and the reference line means field weld, while an open circle at that same junction means weld all around - two different symbols in the same place; and the leg size of a fillet is written to the left of the weld symbol, with length and pitch to its right. Contour symbols show the shape of the finished weld face, and a letter written at the contour symbol names the finishing method - C chipping, G grinding, M machining, R rolling, and others. A note on reading the arrow: the arrow side is the side of the joint the arrow points to - point the arrow at the right side of a joint, and an arrow-side weld goes on the right side.

Q44. Answer: D

Argon and CO_2 are heavier than air and can accumulate at floor level in low-lying areas, displacing oxygen and causing asphyxiation.

Q45. Answer: A

Green is pure tungsten, classification EWP: tungsten makers' charts for the AWS A5.12 classifications band EWP green, at 99.5% tungsten minimum with no oxide addition. Pure tungsten is used on alternating current for aluminum and magnesium, where it balls easily. The other three colours in this question belong to other electrodes: grey is 2% ceriated (EWCe-2), red is 2% thoriated (EWTh-2), and gold is 1.5% lanthanated (EWLa-1.5). Ceriated is the one to watch in the rod drawer: makers now band EWCe-2 grey and note that it was formerly orange, so older stock and older wall charts may still show orange for ceriated - trust the band on the electrode and the maker's current chart.

Q46. Answer: B

CTWD (Contact Tip to Work Distance) is the distance from the contact tip to the workpiece. It affects electrical stickout and therefore current/deposition rate.

Q47. Answer: D

E71T-1 is a rutile-type flux-cored wire used with CO_2 or Ar/CO_2 shielding gas. It produces a fluid, easily removed slag and is suitable for all-position welding.

Q48. Answer: A

The Charpy test strikes a notched specimen with a pendulum and measures the energy absorbed before fracture (in joules). It evaluates toughness at specified temperatures, critical for pressure vessels and low-temperature service.

Q49. Answer: C

In manual CAC-A, **travel speed controls gouge depth**: the faster you travel, the shallower the gouge, and a slower travel speed produces a deeper one. Speeding up does the opposite and leaves the groove even shallower. Pulling the electrode back is wrong. Once the arc is struck you hold a short arc and keep moving fast enough to keep up with the metal removal. If the gap opens too far, the arc goes out. The air jet does not set depth. It blows the molten metal and slag out, and it should point along the gouge. Aiming it at one side only changes which edge the slag sticks to. Electrode diameter sets the groove width, and a groove deeper than about 1-1/2 times the electrode diameter has to be made in more than one pass.

Q50. Answer: B

WHMIS (Workplace Hazardous Materials Information System) is Canada's national hazard communication standard for hazardous products.

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